

Information Security

(CS3002)

Student Name _____ Roll No _____ Section _____ Student Signature _____

Date: September 23, 2024

Course Instructor(s)

Dr. Sufian, Dr. Aqsa, Dr. Nausheen, Dr. Farrukh,
Mr. Usman, Ms. Hira, Ms. Yusra

Sessional-I Exam

SOLUTION

[05 marks]

Total Time: 1 Hours

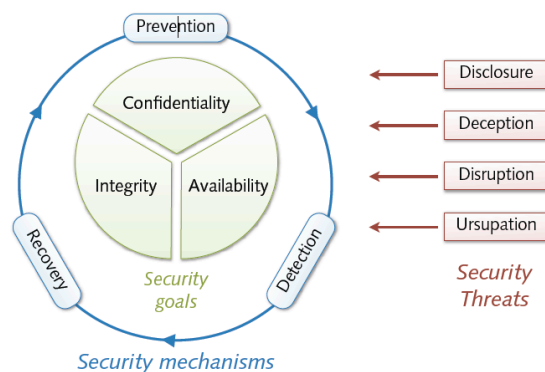
Total Marks: 50

Total Questions: 05

Semester: FA-2024

Campus: Karachi

Dept: CS/SE/AI



Q1b: Draw the following table in the same order in your exam copy and tick in the appropriate column where the given security safeguards apply. [05 marks]

Security Safeguard	Confidentiality	Availability	Integrity
Redundancy		X	
Encryption	X		X
Biometrics		X	
Data Validation Check			X
Secondary Internet Link		X	

CLO # 3: Analyze and apply various security and risk management tools for achieving information security and privacy.

Q2a. Discuss classical cryptographic techniques such as the Caesar cipher and Vigenere cipher. What are their strengths and weaknesses compared to modern cryptographic methods? How do classical cryptographic techniques provide foundational knowledge for understanding modern cryptography?

[05 marks]

Classical Cryptographic Techniques: Caesar and Vigenère Ciphers

Caesar Cipher

The Caesar cipher is a substitution cipher where each letter in the plaintext is shifted a fixed number of places down the alphabet (e.g., a shift of 3 transforms "A" to "D").

Strengths:

Simple to implement and understand.

Fast encryption and decryption processes.

Weaknesses:

Vulnerable to frequency analysis; since there are only 25 possible shifts, it can be easily brute-forced.

Limited security; once the method is recognized, the cipher can be easily broken.

Vigenère Cipher

The Vigenère cipher is a more complex substitution cipher that uses a keyword to determine the shift for each letter, applying a different shift to each letter based on its position in the keyword.

Strengths:

More secure than the Caesar cipher due to the use of multiple shifting values, making frequency analysis more challenging.

Greater complexity increases the difficulty of deciphering without knowledge of the keyword.

Weaknesses:

Vulnerable to Kasiski examination and other forms of frequency analysis if the keyword is short relative to the plaintext.

The security of the cipher relies heavily on the secrecy and randomness of the keyword.

Comparison to Modern Cryptographic Methods

Strengths of Modern Methods:

Algorithmic Complexity: Modern ciphers (e.g., AES, RSA) use advanced mathematical principles that are significantly more secure against various attacks compared to classical techniques.

Key Length: Modern cryptography allows for much longer keys, greatly enhancing security.

Robustness Against Attacks: Modern algorithms incorporate techniques to defend against known cryptanalytic attacks, such as differential and linear cryptanalysis.

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Weaknesses of Classical Methods:

Limited Security: Classical techniques offer minimal security by today's standards, making them unsuitable for protecting sensitive information.

Easily Breakable: With computational resources available today, classical ciphers can be quickly broken.

Foundational Knowledge for Modern Cryptography

Understanding Basic Concepts:

Classical techniques introduce fundamental concepts such as substitution and transposition, which are foundational to more complex modern algorithms.

Historical Context:

Studying classical ciphers helps illustrate the evolution of cryptography, highlighting the importance of mathematical rigor and algorithmic complexity in contemporary methods.

Inspiration for Modern Algorithms:

Many modern encryption techniques are built on principles derived from classical methods, demonstrating how foundational ideas can evolve into sophisticated cryptographic systems.

Q2b. Discuss the characteristics that make stream ciphers suitable for real-time applications. Also brief the security risks associated with using a stream cipher. How does the reuse of keys or initialization vectors compromise its integrity?

[05 marks]

Solution:

Characteristics of Stream Ciphers for Real-Time Applications

Speed and Efficiency:

Stream ciphers process data one bit or byte at a time, making them faster than block ciphers, which operate on fixed-size blocks. This efficiency is crucial for real-time applications such as video conferencing and online gaming.

Low Latency:

The ability to encrypt and decrypt data in small increments minimizes delays, allowing for seamless data transmission. This is essential for applications requiring immediate feedback.

Variable-Length Input:

Stream ciphers can handle inputs of arbitrary length, making them flexible for various data streams, such as audio or video streams, without needing padding.

Simplicity of Implementation:

Stream ciphers often have simpler algorithms compared to block ciphers, which can lead to easier integration in systems with limited processing power or memory.

Security Risks Associated with Stream Ciphers

Key Reuse Vulnerability:

Using the same key or initialization vector (IV) for multiple sessions can lead to serious security flaws. If an attacker captures multiple ciphertexts encrypted with the same key, they can perform statistical analysis to derive the key or plaintext.

IV Management:

The security of stream ciphers heavily relies on the proper management of IVs. If an IV is predictable or reused, it can enable attacks such as bit-flipping or plaintext recovery.

Exposure to Known-Plaintext Attacks:

If an attacker knows or can guess parts of the plaintext, they may exploit this knowledge to derive the key, especially if the same key is used for multiple encryptions.

Impact of Key or IV Reuse on Integrity

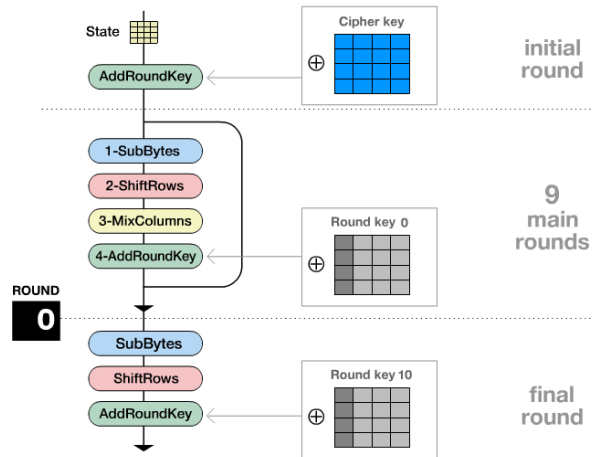
Compromised Confidentiality:

Reusing keys or IVs allows attackers to identify patterns in the encrypted data. For instance, if two identical plaintexts are encrypted with the same key, the resulting ciphertexts will also be identical. This predictability can lead to attacks where the attacker can infer information about the plaintext.

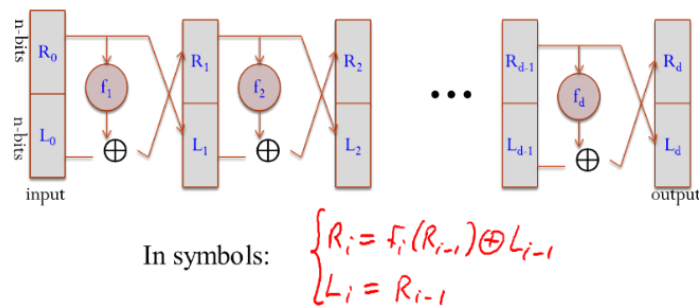
CLO # 3: Analyze and apply various security and risk management tools for achieving information security and privacy.

Q3a: Outline and illustrate the key steps of AES Encryption process.

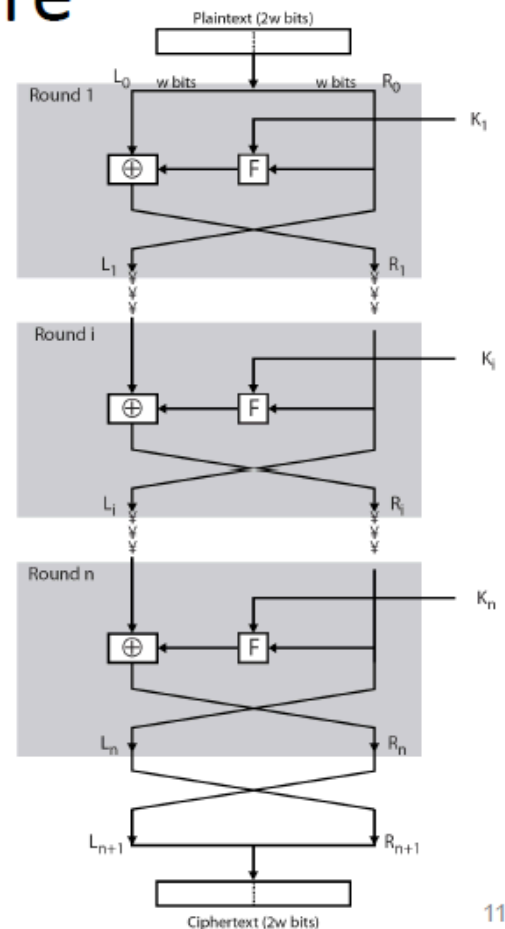
[05 marks]



Q3b: DES is based on Feistel Network which is an invertible function such that $F: \{0,1\}^{2n} \rightarrow \{0,1\}^{2n}$. Draw the Feistel function or write the function in Symbols. **[05 marks]**



Feistel Cipher Structure



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CLO # 4: Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security.

Q4a: Write the encryption and decryption function of ECB and CBC modes.

[05 marks]

SOLUTION

ECB:

Encryption: $C_i = E_K(P_i)$

Decryption: $P_i = D_K(C_i)$

CBC:

Encryption: $C_i = E_K(P_i \oplus C_{i-1}), C_0 = IV$

Decryption: $P_i = D_K(C_i) \oplus C_{i-1}, C_0 = IV.$

Q4b: During the transmission of a DES encoded messages a single-bit error occurs. How many of the bits of the decrypted cipher text can be distorted at most, i) ECB ii) CBC

[05 marks]

SOLUTION

- Using ECB each block is en/decrypted independent of other blocks. If an error occurs in the cipher text block c_i , this error affects only the plain text block m_i , i.e. a maximum of 64 bit may be corrupted.
- Using CBC for decryption the cipher text block c_i is used in the calculation of the plain text block m_i as well as m_{i+1} . In order to calculate m_{i+1} , c_{i+1} is decrypted first and is then combined with c_i . A 1 bit error in c_i may result in a single bit error in m_{i+1} . Hence a maximum of 65 bit may be corrupted, 64 bit in m_i and 1 bit in m_{i+1} (maximum 2 blocks).

CLO # 4: Identify appropriate techniques to tackle and solve problems of real life in the discipline of information security.

Q5a: Alice wants to send a message to Bob. Alice wants Bob to be able to ensure that the message did not alter on a channel. Outline or illustrate the steps that Alice and Bob must follow to ensure the integrity of the message. [05 marks]

Solution:

- Alice (Sender)
 - i) Generates message M
 - ii) Generates $MAC = h(M, K)$ with M and K as input parameters
 - iii) Sends $\{M, MAC\}$ to Bob
- Bob (Receiver)
 - i) Receives $\{M', MAC\}$ (message denoted as M')
 - ii) Generates $MAC' = h(M', K)$ from M' .
 - iii) Compares MAC' and MAC
 - iv) If $MAC = MAC'$ then Bob knows the message was unchanged in transit

Q5b: There are two major constructions of MACs i.e. HMAC (hash based MAC) and MAC generated from a secure PRF (pseudo-random function). What is the minimum security requirement of the tag (or digest) size in bits for each of the two MAC constructions? What seems to be the reason behind the difference in the required digest bits for the two MAC constructions? [05 marks]

SOLUTION

A) PRF MACs $\square$ 80 bits

HMACs $\square$ 160 bits

B) HMACs and PRF-MAC have different security considerations. If the MAC is based on secured PRF, the MAC needs to be secure against exhaustive search and this means that the digest size must be greater than or equal to 80 bits.

In HMAC the security is based on the collision resistance of the HMAC functions. Due to Birthday paradox the attack time is reduced to $O(2^{n/2})$. Therefore, the digest size must be greater than or equal to 160 bits.